

MTH 209: Data Science Lab II (2025-2026 - Sem II)

INSTRUCTOR	Dootika Vats Faculty Building 580	<i>E-mail:</i> dootika@iitk.ac.in <i>Web:</i> https://dvats.github.io/
COURSE DESCRIPTION	The aim of the course is to give students hands-on exposure to various computational and implementation issues in fundamental statistics and probability. There aren't too many unifying themes and the course is made of distinct weeks. Each week will have one unifying theme. Special focus will be on writing reproducible codes, project reports, efficiency, and visualizations. Please note that the schedule below is tentative and may be subject to change. • Week 1: Jan 6: Matrix decompositions and stable operations • Week 2: Jan 13: More matrix and coding practices – Led by TAs • Week 3: Jan 20: Numerical optimization • Week 4: Jan 27: Numerical optimization • Week 5: Feb 3: Parallel computing • Week 6: Feb 10: Simulations studies • Week 7: Feb 17: Fun topic 1 • Week 8: Feb 24: Midsem week • Week 9: Mar 3: Midsem break • Week 10: Mar 10: Covariance matrices and PCA • Week 11: Mar 17: Covariance matrices and PCA – Led by TAs • Week 12: Mar 24: Dimension reduction • Week 13: Mar 31: Fun topic 2 • Week 14: Apr 7: Presentations – Online • Week 15: Apr 14: Presentations – Online • Week 16: Apr 21: Fun topic 3	
PREREQUISITES	MTH208: Data Science Lab I	
SCHEDULE	<u>Location:</u> L05 for Lecture and Room 301, New Core Lab (CC02) for Lab <u>Timings:</u> Lecture: Tuesdays, 5:10 pm - 6:00 pm, Lab: Wednesdays and Thursdays: 2:00pm - 3:00pm	
STRUCTURE	This is a lab course with 1 lecture and 2 labs hours. The theory for the week will be discussed in the lectures and the labs will cover practical implementations and in-lab assignments. Labs will be worksheet focused; a worksheet will be given in the lab and is expected to be completed during the lab hours. The worksheet will be mostly self-sufficient, but often I will explain solutions, concepts, and key steps. The goal is to not take anything home to do.	
COURSE WEBPAGE AND GITHUB	The main resource page for the course is the course on HelloIITK. Additionally, we will be using GitHub Classroom for lab worksheets, assignment submissions etc. <u>Make sure you have a GitHub account before the first day of class.</u>	
PROJECT	A significant component of the course will be a project. More information on the project will be given in due course.	

ATTENDANCE Attendance is not compulsory. However, there will be an assignment every week of the lab.

MARKS
DISTRIBUTION The following are the marks distribution

Assignments	30%
Project	20%
Mid-sem Exam	20%
Final Exam	30%

The final grade will be relative and the minimum assignment for every student will be dropped. All exams will happen in the lab.

ACADEMIC
HONESTY IIT Kanpur is committed to stop the use of unfair means in academic activities. The following policy will be followed in this course:

Use of unfair means in course-work: If a student cheats or uses any unfair means in the course:

1. The student will get an F in the course.
2. The student will not be allowed to drop the course.
3. The name of the student will be made public to the students of the course.
4. The matter will be sent to the respective departments' ethics/student affairs' committee.
5. A complaint will be sent to SSAC for disciplinary action.
6. An intimation will be sent to DoAA to be kept in the student's record.

What constitutes cheating:

1. Copying code from each other during assignments/exams
2. Copying any part of the code for GPT systems
3. Using plagiarised language in any report. Using AI for the purpose of writing reports constitutes plagiarism.
4. Unapproved accessing of the internet during assignment/exams.
5. If you are found talking even for a brief moment in an assignment/exam, you will be asked to leave immediately.

REFERENCES There is no main text for this book, but the following references may be helpful:

- Student-sourced book to prepare for this course:
<https://dvats.github.io/BasicRProblems/>
- Wickham, Hadley. "Elegant graphics for data analysis." O'Reilly Media 35.211 (2009)
- Wickham, Hadley. "Mastering shiny". O'Reilly Media, Inc., 2021.
- Bruce, Peter, Andrew Bruce, and Peter Gedeck. "Practical statistics for data scientists: 50+ essential concepts using R and Python". O'Reilly Media, 2020.
- VanderPlas, Jake. "Python data science handbook: Essential tools for working with data." O'Reilly Media, Inc., 2016.
- Boehmke, Bradley C. "Data wrangling with R". Springer International Publishing, 2016.
- Pineau, Joelle, et al. "Improving reproducibility in machine learning research (a report from the NeurIPS 2019 reproducibility program)." Journal of Machine Learning Research 22 (2021).